

YASHAS BESANAHALLI VASUDEVA

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Education

University of North Carolina at Charlotte (UNCC)

Expected May 2025

Master of Science in Computer Science

Charlotte, NC, USA

Siddaganga Institute of Technology

Aug. 2018 – Aug. 2022

Bachelor of Engineering in Computer Science Engineering

Tumkur, Karnataka, India

Professional Experience

University of North Carolina at Charlotte

Jan 2025 - Present

Graduate Research Assistant | AI, Swift, Python, Pytorch

Charlotte, NC, USA

- I am actively collaborating with the AI and iOS development team on innovative, upcoming AI-driven projects under the guidance of Professor Dr. Pu Wang. This opportunity allows me to contribute to cutting-edge advancements in AI, Deep Learning and LLM while leveraging my skills in Swift (iOS development) to create impactful solutions.
- Leading the Wav2Lip: Lip Sync in the Wild project, I am spearheading feature enhancements as we prepare to launch an iOS app with realtime streaming using Coreml model on the App Store in the coming days.

Mavenir

May 2024 – August 2024

Software Engineer Intern | Python, Pytest, Flask, Kubernetes

Richardson, TX 75081, USA

- Worked on 5G Radios (DU Emulator Radio, which was made for the dish network) where I have complete knowledge about RF transceiver, RU, DU, CU, Fronthaul PTP switch diagnostics, SFP light levels, RET tests, GNSS and GPS lock status etc.
- Implemented and deployed new features for Mavenir's MAVDue product, including API-based execution and application containerization, improving system scalability and reducing deployment time by 30%.
- Designed an automated regression testing environment with Pytest (85% test coverage) and optimized MAVDue application interfaces, improving user efficiency by 20%.

Oracle Cerner

Sep 2022 – March 2023

Software Engineer 1 | Clojure, Oracle Apex

Bengaluru, Karnataka, India

- Worked as a Backend Developer, gaining hands-on experience in Clojure through training provided by Synapse Academy. Successfully completed tasks while writing and validating efficient test cases to ensure robust functionality.
- Implemented and tested final project deliverables using Jenkins.
- Developed web applications using Oracle Apex, managing patient data and improving insights into patient performance, leading to a 25% enhancement in data accessibility and analysis efficiency.

Oracle Cerner

March 2022 – Sep 2022

Software Intern | Ruby on Rails

Bengaluru, Karnataka, India

- Worked as a Full-Stack Developer on the "Algo-Builder" project, implementing and managing CRUD operations using Ruby on Rails to enhance system functionality and scalability.
- Utilized Postman to validate API responses, ensuring 100% accuracy in output and seamless integration.
- Leveraged Jenkins, Rubocop, and RSpec to maintain code quality, enforce proper indentation, and achieve 95% test case coverage, ensuring reliable and efficient application performance.

Projects

AWS Automation Pipeline | AWS, Pyspark, Sagemaker, Quicksight, Lambda

December 2024

- Engineered a dynamic S3 bucket and EC2 instance to seamlessly store and manipulate uploaded data, ensuring efficient handling and scalability. Leveraged Amazon SageMaker to develop and deploy advanced machine learning models, achieving 95% prediction accuracy on key metrics.
- Harnessed Amazon QuickSight to visualize machine learning results, analyzing over 1 million data points and creating 10+ insightful dashboards.
- Fully automated the entire workflow using AWS Lambda functions, optimizing processes and enhancing operational efficiency.

Wav2Lip : Accurately Lip-synching Videos In the Wild. | Python, Flask, Flutter, Dart, Xcode, Swift

November 2024

- Developed a web application using Flask and a mobile application with Flutter and Dart for Wav2Lip, a GAN-based deep learning model that enables realistic lip synchronization by aligning lip movements in videos with audio files, improving multimedia functionality and user experience.
- Converted the Wav2Lip PyTorch model to Core ML, enabling seamless integration on iPhones and enhancing its applicability for video dubbing, virtual assistants, and video conferencing.

Mobile Travel Tutor | Flutter, Dart, MongoDB

August 2022

- Designed and developed an Android-based Tourist Guide System focused on enhancing travel experiences in the Karnataka region, integrating features like GPS navigation, place query information, and multi-output functionality, improving trip planning efficiency by 35% and user engagement by 40%.

Technical Skills

Languages: C++, Python, HTML5/CSS3, Python, SQL, Ruby on Rails, Dart, Java, Swift

Tools: Git, VS Code, Postman, MongoDB Atlas, MySQL Workbench, Docker, Kubernetes, Photoshop, Flask, Pytest, Pyspark, Xcode, Pytorch, Coreml, ONNX

Technologies: React, SQL Server, AWS, Google Cloud, Azure Databricks, Delta Lake, HuggingFace, Tableau

Development Practices: Test-Driven Development (TDD), Agile, Responsive Design, Code Optimization

Leadership / Certifications/ Extracurricular

- Achieved the honor of being named the **second-best project award** in the entire graduate class for the development of the wav2lip web application and iOS app development.
- Team Lead for the wav2lip project, focusing on AI and iOS development and training other graduate students.
- **AWS Academy Graduate – Data Engineering (Dec 2024)** 